Project Proposal: Architecting Autonomous AI – A Comparative Analysis of Agentic Design Patterns

Yiming Wang (7750857)
Yucheng Tu (8699902)
Kapila Wijetunge (6031067)

October 20, 2025

Abstract

Autonomous agentic AI redefines large language models (LLMs) from passive predictors into goal-oriented systems capable of planning, acting, and self-correction. Existing studies commonly benchmark a few design patterns—Reflex, ReAct, Chain-of-Thought (CoT), and Tree-of-Thought (ToT)—mainly on task accuracy or latency, while overlooking key aspects such as reasoning quality, action fidelity, robustness, transparency, controllability, hallucination risk, and safety. This project proposes a pattern-agnostic evaluation framework structured across three analytical layers—Cognitive, Behavioral, and Systemic—encompassing seven dimensions: reasoning quality; cognitive safety and constraint adherence; action–decision alignment; success and efficiency; behavioral safety; robustness and scalability; and an integrated metric for controllability, transparency, and resource efficiency. Proof-of-concept agents will be implemented in LangGraph through lightweight adapters that standardize the think, act, observe cycle and generate unified logs for comparative analysis. Evaluation metrics will be normalized and aggregated to produce dimension-level and composite scores, with sensitivity analysis ensuring robustness across task contexts. Expected contributions include: (1) a reusable, pattern-independent evaluation protocol; (2) standardized agentic implementations with unified telemetry; and (3) quantitative analyses of trade-offs—such as reasoning depth versus efficiency and robustness versus computational cost—to inform future architecture selection.

Keywords: Agentic AI; Large Language Models (LLMs); Evaluation Framework; Reasoning Quality; Cognitive and Behavioral Safety; Robustness; Controllability; LangGraph.

1 Introduction

Large language models (LLMs) have evolved rapidly from passive text predictors into autonomous agents capable of planning, acting, and adapting through feedback. Emerging paradigms such as Reflex, ReAct, Chain-of-Thought (CoT), and Tree-of-Thought (ToT) have shown that different reasoning and control strategies can significantly affect performance and interpretability. However, despite these advances, the evaluation of agentic systems has not progressed at the same pace. Existing studies often focus on task accuracy or latency while overlooking deeper dimensions such as reasoning quality, robustness, controllability, and ethical safety [1]. As a result, there remains limited understanding of how different design patterns compare under consistent and transparent conditions.

This project addresses that limitation by developing a systematic, pattern independent framework for evaluating agentic architectures. The framework analyses agent performance through 3 complementary levels, cognitive, behavioural, and systemic, covering aspects of reasoning, decision execution, and overall stability. Within these levels, 7 measurable dimensions capture key properties including reasoning coherence, efficiency, robustness, transparency, and safety. The design aims to unify metrics and logging standards to enable consistent comparison and reproducible experimentation across diverse agentic models.

Ultimately, this study seeks to establish a transparent and fair benchmark for evaluating autonomous artificial intelligence systems. By promoting methodological consistency and interpretability, the proposed framework contributes to more reliable assessment practices and provides guidance for the future development of agentic LLM architectures.

1.1 Aims

1.1.1 Overall Aim

This project aims to design and validate an evaluation framework that allows fair comparison among agentic design patterns while maintaining independence from any single architecture. It seeks to capture both quantitative and qualitative aspects of performance, thereby clarifying the trade-offs between reasoning depth, operational efficiency, and system stability.

1.1.2 Specific Objectives

- **Framework Design:** Develop a structured model that integrates cognitive, behavioural, and systemic dimensions with clearly defined metrics and scoring procedures.
- **Implementation and Reproducibility:** Build lightweight adapters in LangGraph to standardise the think–act–observe cycle and enable consistent experiment logging and replication.
- **Benchmark Construction:** Create a balanced suite of reasoning, planning, and safety-oriented tasks to evaluate representative agentic patterns such as Reflex, ReAct, CoT, and ToT.

- **Quantitative Analysis and Guidance:** Examine performance trade-offs (e.g., reasoning depth versus efficiency, robustness versus cost) and provide practical recommendations for architecture selection.

Expected deliverables include an open evaluation protocol, reference implementation, benchmark dataset, and a technical report presenting experimental findings and guidance for future research.

1.2 Related Work

Research on agentic LLMs has advanced significantly, demonstrating diverse ways of combining reasoning and action to achieve autonomy. Early frameworks such as CoT and ToT reveal that structured reasoning can improve interpretability and accuracy, though often at the cost of higher computational demand [1]. The ReAct paradigm interleaves reasoning and action to enhance contextual grounding but increases token consumption and error sensitivity [2]. More recent developments, including Plan-and-Execute and Reflexion, incorporate memory and self-reflection mechanisms to support long-horizon reasoning, yet their evaluations remain mostly qualitative [4].

Current evaluation practices emphasise task success, efficiency, and safety compliance. Large-scale benchmarks such as AGENTIF [6] and AgentQuest [7] provide quantitative insights into instruction-following and general problem-solving but suffer from inconsistent metrics and logging conventions. Targeted studies like MME-CoT [1] focus on reasoning quality and robustness but remain limited to cognitive evaluation. Surveys indicate that system-level attributes—transparency, controllability, and scalability—are still rarely addressed [3].

Despite these contributions, the research landscape remains fragmented. Most studies measure how well agents perform tasks without examining whether their reasoning aligns with their actions or whether they maintain stability under changing conditions.Wicaksono et al. [5] highlight that such misalignments can expose vulnerabilities, calling for structural evaluation through action graphs. Safety-focused benchmarks [8, 15] have made progress in assessing ethical adherence but rarely integrate these insights with performance metrics. Likewise, mobile and embodied contexts introduce new challenges related to perception, task diversity, and interaction [9].

In response, this study contributes a unified evaluation paradigm that consolidates these fragmented efforts into a coherent methodological foundation. By integrating cognitive, behavioural, and systemic perspectives under consistent principles, and by adopting standardised telemetry and data pipelines through LangGraph, the framework advances methodological rigour and ensures transparent, reproducible assessment of autonomous LLM agents.

2 Methodology

2.1 Background and Motivation

Recent progress in agentic artificial intelligence has resulted in a variety of design patterns, including Reflex, ReAct, Chain of Thought (CoT) and Tree of Thought (ToT), that convert large language models from passive predictors into interactive agents capable of planning, acting and adapting. Although these architectures differ in reasoning depth and control strategy, evaluation practices often continue to emphasise task accuracy or computational efficiency, thereby overlooking essential properties such as reasoning soundness, controllability, robustness and ethical reliability. As study by [3], benchmark frameworks for agentic LLMs remain fragmented and lack consistent criteria for fairness and reproducibility.

To bridge this gap, this study proposes a multi-level evaluation framework that analyses agentic architectures through three interconnected lenses: cognitive, behavioural and systemic (Table 1). The cognitive level addresses internal reasoning processes and adherence to logical and ethical constraints; the behavioural level considers how decision-making translates into observable actions, success rates and operational efficiency under realistic conditions; the systemic level examines overall system behaviour, including robustness under perturbation, transparency of operations, scalability and resource governance [4, 5].

By integrating these seven evaluation dimensions into one coherent structure, the framework enables a holistic view of how reasoning, decision-making and system integrity interact—thereby advancing methodological rigour and ethical accountability in agent evaluation [6, 11].

Layer	**Evaluation Dimensions**	**Purpose / Focus**	**Key References**
Cognitive	(1) Reasoning Quality (2) Safety & Constraint Adherence (Cognitive)	Assess logical coherence, depth of reasoning, and value alignment in thinking	[1, 6]
Behavioural	(1) Action–Decision Alignment (2) Success & Efficiency (3) Safety & Constraint Adherence (Behavioural)	Evaluate consistency between reasoning and observable actions, efficiency, and behavioural safety	[2, 3, 6]
Systemic	(1) Robustness & Scalability (2) Controllability,Transparency & Resource Efficiency	Review stability, transparency, and resource efficiency at system scale	[4, 5]

Table 1: Evaluation Framework Overview

2.1.1 The Rationale for the Cognitive–Behavioural–Systemic Framework

The use of a hierarchical evaluation framework allows clear differentiation between an agent's internal reasoning, its external actions and its system-level performance. Without this separation, reasoning quality may be conflated with execution speed or system stability. Prior work emphasises multi-level evaluation, for example, Mohammadi et al. [11] show that evaluation frameworks should reflect both what to evaluate (capabilities, safety, reliability) and how to evaluate (interaction modes, metrics, tooling). The proposed tripartite model aligns with this insight and mirrors the architecture of many agentic systems, which can be decomposed into internal reasoning (thought generation), action execution (tool use and decision making) and runtime management (resources, logging, feedback). Thus, this framework is both scalable and broadly applicable across heterogeneous agent designs.

2.2 Implementation method & Proxy indicators

To operationalise the proposed three-layer, seven-dimension evaluation framework, this study adopts a structured experimental methodology that integrates both direct measurement and proxy estimation. Each evaluation dimension is defined through measurable sub-indicators, with normalisation and aggregation procedures designed to ensure comparability and reproducibility across diverse agentic patterns.

The overall scoring procedure follows a three-step principle: (1) each sub-indicator is normalised to the 0–1 range; (2) dimension-level scores are obtained by averaging the sub-indicators; and (3) composite results are computed using either uniform or weighted combinations. A sensitivity analysis of weight variation is conducted in the appendix to examine robustness of conclusions. This design allows interpretability without sacrificing methodological rigour [3, 12]

Implementation relies on LangGraph, which serves as a lightweight adapter providing a unified think–act–observer cycle and a standardized telemetry schema (including thought, action, observation, token usage, latency, policy_flags and termination states). Prototype implementations of 4 representative patterns, Reflex, ReAct, CoT, and ToT, have been developed for experimental testing. Stage 1 emphasises framework validation and metric calibration; Stage 2 extends to large-scale comparative analysis, replacing proxy indicators with validated quantitative measures. Similar staged evaluation has been successfully applied in prior benchmark frameworks such as AgentQuest [7], AgentBoard [12] and [13].

Where ambiguity persists, selective human validation will be conducted to verify the correctness of reasoning and ensure interpretive reliability, consistent with hybrid verification approaches discussed in recent human-centered evaluation frameworks [17].

2.2.1 Cognitive Layer – Evaluating the Thinking Process

The cognitive layer investigates how the agent reasons internally and whether its reasoning process is logically coherent, accurate and ethically aligned.

(1) **Reasoning Quality.** This dimension assesses the internal coherence and correctness of the agent's reasoning trace. The analysis proceeds by extracting reasoning sequences from structured logs and evalu-

ating them through either (i) automated judgement using an external large model prompted for coherence scoring, or (ii) self-consistency analysis in which multiple runs are compared through embedding similarity and step-level alignment [1]. Where gold-standard intermediate steps exist, structural alignment via Levenshtein distance is performed to compute step accuracy. In cases without annotated references, final-answer agreement or self-consistency majority serves as a validated proxy indicator.

(2) **Cognitive Safety and Constraint Adherence.** This dimension examines whether the reasoning process exhibits harmful bias, factual inconsistency or ethical violations. Automated filters perform lexical and semantic screening for toxicity or unsupported claims [14, 15]. For high-risk cases, a secondary classifier or a judge-LLM is applied to verify factual grounding. In Stage 1, where comprehensive retrieval verification is not yet implemented, keyword-based anomaly detection and heuristic scoring are used as proxies to flag reasoning segments for manual audit in Stage 2.

A particular focus within cognitive safety is the detection of hallucination, defined as the generation of unsupported or factually inconsistent content. Following recent evaluation protocols [1, 14–16], hallucination frequency will be measured as the proportion of reasoning traces containing unverifiable claims or contradictions to retrieved evidence.

2.2.2 Behavioural Layer – Evaluating Actions and Execution

The behavioural layer evaluates how well the agent's actions align with its internal reasoning, the efficiency of task execution and the safety of observable behaviour.

(3) **Action–Decision Alignment.** This dimension measures the correspondence between an agent's stated intention and the action actually performed. Semantic comparison is conducted between planned and executed actions using verb–tool mappings and embedding-based similarity. For early experimental stages, the analysis can be simplified through explicit plan strings ("Next I will ...") and string-level matching, which serves as an interpretable proxy for alignment quality [2].

(4) **Success and Efficiency.** This dimension quantifies whether tasks are completed successfully and how efficiently the agent operates. Each run is automatically labelled with a binary success outcome and associated resource costs, including total tokens, execution time and API latency. Efficiency is expressed as a normalised cost score combining token and time usage, providing a scalable comparison across architectures [6, 13]. When detailed cost metrics are unavailable, the ratio of actual step count to an upper-bound budget acts as a proxy indicator for operational efficiency.

(5) **Behavioural Safety.** This metric ensures that agent actions conform to predefined safety and compliance requirements. Tool invocation is validated through whitelisted APIs, domain-restricted parameters and controlled execution environments. Violation rates, blocked attempts and sandbox logs form quantitative evidence of behavioural adherence. In preliminary experiments, static domain regex checks provide a lightweight proxy until full sandbox instrumentation becomes available [8, 14].

2.2.3 Systemic Layer – Evaluating Operational Integrity

The systemic layer examines the broader operational characteristics of the agent system—robustness, scalability, transparency, controllability and resource efficiency.

(6) **Robustness and Scalability.** This dimension evaluates how consistently the agent performs under stochastic or environmental perturbations. Controlled temperature and seed sweeps are conducted to compute variance-based stability indices. Perturbation tests introduce input noise (e.g., paraphrases or typos via nlpaug or textattack) to measure degradation: $\Delta = |S_{clean} - S_{noisy}|$. Progressive scaling experiments assess the slope of performance decline as task complexity increases. For Stage 1, variance based stability alone is used as a proxy for full perturbation robustness [5].

(7) **Controllability, Transparency and Resource Efficiency.** This composite dimension reflects whether the system allows human intervention, maintains interpretable logs and manages resources effectively.

- Controllability is measured through runtime override tests (stop/redo success rate).
- Transparency is quantified by trace completeness, the proportion of steps containing full thought–action–observation records, and supplemented by human or judge-LLM clarity assessments.
- Resource Efficiency reuses normalised token/time costs from the efficiency metric. Where direct human evaluation is infeasible, automated trace completeness and policy-flag frequencies serve as proxy measures for transparency and controllability [4, 11].

Summary Together, these implementation procedures transform conceptual evaluation principles into a reproducible experimental pipeline that integrates automated scoring, statistical validation and structured proxies. By aligning quantitative measurement with interpretability and safety auditing, the proposed methodology enables a fair, transparent and scalable assessment of heterogeneous agentic AI architectures [3, 12].

2.3 Project Challenges

While the framework provides a structured pathway for cross-pattern evaluation, several key challenges are anticipated (see Table 2). These include fairness across heterogeneous patterns, measurement of unavailable metrics, cost and latency constraints, reproducibility concerns, safety and compliance, scope control and resource bottlenecks. Our mitigation strategies include standardised adapters, proxy metrics substitution and replacement, strict budget controls, repeated trials and public release of artefacts, policy filtering for safety, staged protocol freeze and early-stop rules for slow runs.

Challenge	Why it matters	Mitigation Strategy
C1. Fair comparison across heterogeneous patterns	Different agentic patterns expose diverse reasoning loops and log structures, making direct score comparison difficult.	Design a lightweight Adapter API that standardises the think–act–observe interface and logging schema across all patterns.
C2. Availability of some metrics in Stage 1	Some indicators may not be fully measurable in early phases.	Use validated proxy metrics and upgrade them in Stage 2.
C3. Cost and latency management	Long decision loops and large token counts can cause budget overrun and disrupt repeated trials.	Enforce strict token/time limits, pilot calibration, cache tool results and monitor cost per run.
C4. Reproducibility and statistical stability	Unstable results or inconsistent configurations undermine credibility.	Fix model versions and seeds; 3–5 repeats; publish logs; report mean $\pm$ 95% CI.
C5. Safety and compliance concerns	Unsafe prompts or tool interactions may result in harmful outputs or policy violations.	Whitelist tools/data; automated policy–flagging; human review; abort on confirmed violation.
C6. Scope and timeline management	Over-broad scope may compromise deliverables and analysis quality.	Freeze protocol after pilot; expand only with supervisor approval; keep a three–week buffer.
C7. Slow runs and resource bottlenecks	Some ToT branches or long–horizon tasks may take hours, limiting repeatability and comparison.	Impose step/time budgets (e.g., $\leq$ 30 min/run); early–stop; prune beam; cache/memoize; parallel queues; report time–normalised proxies.

Table 2: Challenges and Mitigation Plans

2.4 Plan and Time Line

The timeline for the project is shown in Figure 1.

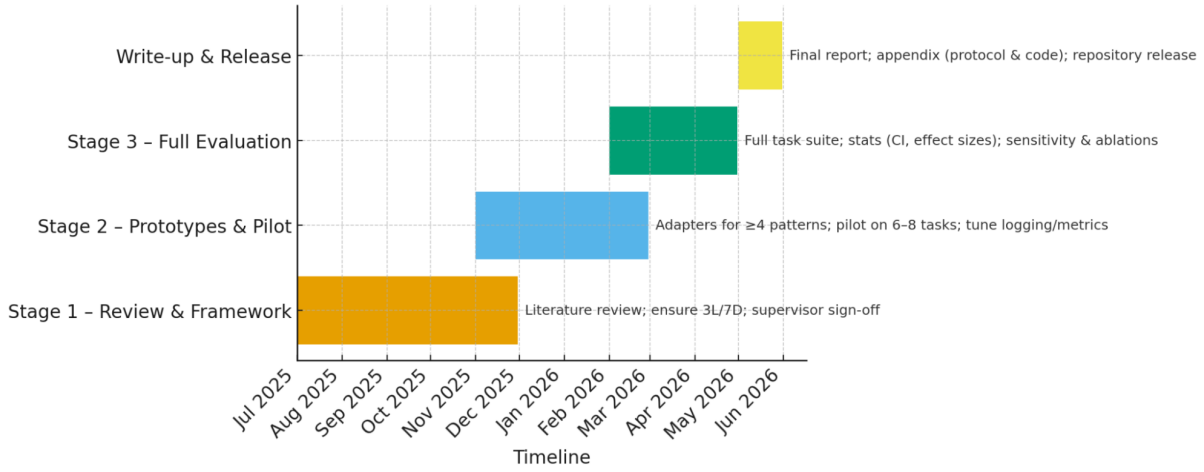

Figure 1: Plan and Time Line

3 Expected Outcomes

This project is structured to deliver foundational clarity on the efficacy and trade-offs of next-generation autonomous agentic AI architecture. The outcomes are specific, measurable, and essential for transitioning from theoretical LLM use to reliable, multi-step agent deployment.

3.1 Experimental Deliverables: A Data-Backed Comparison of Agentic Design Patterns

The core outcome will be a comparative performance dataset, produced by applying a structured evaluation methodology to distinct agentic prototypes. This dataset will provide objective evidence for assessing the suitability of different design patterns for multi-step tasks.

Data will be captured systematically across key performance metrics and analytical dimensions defined in Section 2.1 of the proposal. The evaluation will measure each prototype's performance using clearly defined metrics:

- **Task Completion Rate (Efficacy):** quantifies the agent's ability to complete dynamic, multi-step tasks.
- **Error Handling Capability (Robustness):** measured as recovery percentage, directly assessing system reliability.
- **Operational Efficiency (Cost and Latency):** evaluates resource trade-offs using token counts and execution time, showing how design choices impact cost and speed.

These metrics will be further refined and validated during the next project phase to ensure precise and consistent measurement.

3.2 Foundational Review and Design Pattern Mapping

Before prototype development, the project will deliver a structured review of existing and emerging agentic design patterns. This review will summarize current frameworks, identify key design approaches, and map how different architectures handle planning, memory, communication, and error recovery. The outcome of this stage will be a design pattern map or taxonomy, establishing a clear foundation for prototype implementation and evaluation. It ensures the project builds on current developments while identifying practical gaps for further exploration.

3.3 Evaluation Methodology and Tools

A key project outcome will be the development and documentation of a consistent evaluation methodology for agentic systems. Deliverables will include:

- **A detailed evaluation guide** describing the testing setup, task structures, and performance measures.
- **Reusable scripts or tools** that automate data collection and analysis for each agent prototype.
- **A validation process** to maintain consistency and fairness across all tests.

This framework will ensure the performance comparison in Section 3.1 is transparent, repeatable, and scalable for future work.

3.4 Technical Analytical Synthesis And Final Deliverables

The outcome of this project will also include functional assets and a comprehensive comparison report, directly fulfilling all project requirements. The functional assets will consist of agent prototypes, evaluation scripts and pipelines, and configuration and analysis tools, all designed to support repeatable testing and transparent analysis.

The project's most tangible technical deliverable will be at least four working proof-of-concept prototypes, each representing a distinct agentic design pattern developed using an agentic framework such as LangGraph. These prototypes will perform multi-step, real-world tasks such as:

- information retrieval with feedback loops,
- structured report generation,

- responding to open-ended user prompts, and
- error recovery during tool use.

These implementations are intended to reveal design trade-offs in observable and measurable contexts, providing clear evidence of how different agentic architectures handle complex workflows.

The entire investigation will culminate in a comparative analysis of agentic design patterns, combining both quantitative results and qualitative observations to deliver practical insights into the efficacy, robustness, and trade-offs of the tested agentic systems. This synthesis will form the project's core contribution, bridging theoretical understanding and real-world performance outcomes.

3.5 Expected Benefits Establishing the Nexus, Career Advancement, and Collaborative Soft Skills

The project provides valuable insights that amplify the value of the final deliverables. A critical outcome is to establish the nexus between the theoretical body of knowledge found in available literature review and realities; the final analysis will explicitly use the prototype results to validate, refute, or modify the performance expectations outlined in the initial literature review findings(where available). This linkage transforms the project into a valuable study that tests foundational assumptions found in the literature review with experimental results.

Crucially, the successful completion of this project ensures essential collaborative skill development. The necessity of integrating components requires all members to actively practice knowledge transfer and ensure that shared responsibility with mutual respect drives the project. This process validates the team's ability to navigate differing technical strengths and ensure every member contributes meaningfully to the project's success.

Furthermore, this project delivers exceptional career work samples for team members, a valuable outcome that is envisaged. The authentic, high-impact technical materials, including codebases, raw performance datasets, and the final comparative analysis, showcase a reasonable competency in design, and evaluating autonomous AI systems, directly enhancing future career prospects and opportunities. The entire experience in this project demonstrates the team's technical skills and understanding of the agentic design patterns as well as the team dynamics at play, a vital outcome.

References

[1] Jiang, D, Zhang, R, Guo, Z, Li, Y, Qi, Y, Chen, X, Wang, L, Jin, J, Guo, C, Yan, S, Zhang, B, Fu, C, Gao, P & Li, H 2025, 'MME-CoT: Benchmarking Chain-of-Thought in Large Multimodal Models for Reasoning Quality, Robustness, and Efficiency', preprint.

[2] Bhambri, S, Verma, M & Kambhampati, S 2025, 'Do Think Tags Really Help LLMs Plan? A Critical Evaluation of ReAct-Style Prompting', *Transactions on Machine Learning Research*.

[3] Yehudai, A, Eden, L, Li, A, Uziel, G, Zhao, Y, Bar-Haim, R, Cohan, A & Shmueli-Scheuer, M 2025, 'Survey on Evaluation of LLM-based Agents', preprint.

[4] Xia, B, Lu, Q, Zhu, L, Xing, Z, Zhao, D & Zhang, H 2024, 'Evaluation-Driven Development of LLM Agents: A Process Model and Reference Architecture', preprint.

[5] Wicaksono, I, Wu, Z, Patel, R, King, T, Koshiyama, A & Treleaven, P 2025, 'Mind the Gap: Evaluating Model- and Agentic-Level Vulnerabilities in LLMs with Action Graphs', preprint.

[6] Qi, Y, Peng, H, Wang, X, Xin, A, Liu, Y, Xu, B, Hou, L & Li, J 2025, 'AGENTIF: Benchmarking Instruction Following of Large Language Models in Agentic Scenarios', preprint.

[7] Gioacchini, L, Siracusano, G, Sanvito, D, Gashteovski, K, Friede, D, Bifulco, R & Lawrence, C 2024, 'AgentQuest: A Modular Benchmark Framework to Measure Progress and Improve LLM Agents', preprint.

[8] Potham, R 2025, 'Evaluating LLM Agent Adherence to Hierarchical Safety Principles: A Lightweight Benchmark for Probing Foundational Controllability Components', preprint.

[9] Deng, S, Xu, W, Sun, H, Liu, W, Tan, T, Liu, J, Li, A, Luan, J, Wang, B, Yan, R & Shang, S 2024, 'Mobile-Bench: An Evaluation Benchmark for LLM-based Mobile Agents', preprint.

[10] Hu, X, Zhao, Z, Wei, S, Chai, Z, Ma, Q, Wang, G, Wang, X, Su, J, Xu, J, Zhu, M, Cheng, Y, Yuan, J, Li, J, Kuang, K, Yang, Y, Yang, H & Wu, F 2024, 'InfiAgent-DABench: Evaluating Agents on Data Analysis Tasks', *arXiv preprint* arXiv:2401.05507, available at: `https://arxiv.org/abs/2401.05507`.

[11] Mohammadi, M, Li, Y, Lo, J & Yip, W 2025, 'Evaluation and Benchmarking of LLM Agents: A Survey', in *Proceedings of the International Conference on Knowledge Discovery and Data Mining*, pp. 6129–6139.

[12] Ma, C, Zhang, J, Zhu, Z, Yang, C, Yang, Y, Jin, Y, Lan, Z, Kong, L & He, J 2024, 'AgentBoard: An Analytical Evaluation Board of Multi-turn LLM Agents', preprint.

[13] Anonymous 2025, 'AgentRace: Benchmarking Efficiency in LLM Agent Frameworks', OpenReview preprint.

[14] Yuan, T, He, Z, Dong, L, Wang, Y, Zhao, R, Tian, X, Xu, L, Zhou, B, Li, F, Zhang, Z, Wang, R & Liu, G 2024, 'R-Judge: Benchmarking Safety Risk Awareness for LLM Agents', *arXiv preprint* arXiv:2401.10019v3.

[15] Zhang, Z, Cui, S, Lu, Y, Zhou, J, Yang, J, Wang, H & Huang, M 2024, 'Agent-SafetyBench: Evaluating the Safety of LLM Agents', *arXiv preprint* arXiv:2412.14470v2.

[16] Ji, Z, Lee, N, Frieske, R, Yu, T, Su, D, Xu, Y, Ishii, E, Bang, YJ, Madotto, A & Fung, P 2023, 'Survey of Hallucination in Natural Language Generation', *ACM Computing Surveys*, vol. 55, no. 12.

[17] Sung, YY, Kim, H & Zhang, D 2025, 'VeriLA: A Human-Centered Evaluation Framework for Interpretable Verification of LLM Agent Failures', *arXiv preprint* arXiv:2503.12651.